

Advay Ratan

(628) 255-0206

advayratan@berkeley.edu

advayratan.com

linkedin.com/in/advay-ratan/

Berkeley, CA

EDUCATION

University of California, Berkeley

Aug 2022 - May 2026

B.S. in Electrical Engineering and Computer Science

GPA: 4.0

Relevant Coursework: *Compilers, Quantum Computing, Microfabrication, Tapeout* (A), Digital Design and ICs (A+), Computer Architecture (A), Algorithms & Datastructures (A+), Networks (A), Operating Systems (A), Microelectronic Devices (A+), Discrete Math & Probability (A+), Signals and Systems (A+), Optimization (A), Databases (A+), Analog Design (A), Probability & Random Processes (A), Graphics (A+)

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, Bash, Verilog, System Verilog, VHDL, RISC-V, Tcl, Chisel

Software: Git, Linux, SPICE, KiCad, Cadence Virtuoso, Synopsys PrimeTime, Cadence Innovus

Hardware: FPGA, Arduino, Electrical Lab Equipment, Physical Design, PCB Design, Xilinx Vivado

WORK EXPERIENCE & PROJECTS

SDE Intern | Annapurna Labs

Aug 2025 - Dec 2025

Cupertino, CA

- Built near-cycle-accurate performance model for custom ML accelerator chip used by 4+ teams
- Analyzed micro-arch, Verdi waveforms for <5% error at 1000x performance of RTL simulators

FPGA Intern | Optiver

Jun 2025 - Aug 2025

Chicago, IL

- Added feature to core quoting-side FPGA system deployed in all exchanges
- Wrote multiple new VHDL modules, Verilog testbenches and corresponding C++ software APIs
- Used Vivado to meet aggressive timing requirements via logic optimization and pipelining

CPU Design Intern | Apple

May 2024 - Aug 2024

Cupertino, CA

- Worked on STA methodology improvements and clock tree robustness checks
- Designed Python and TCL scripts from the ground up to interface with tools and parse large text files
- Created parallelized flows to run, queue, parse, and analyze results of thousands of simulation jobs

Undergraduate Researcher | Berkeley SLICE Lab

Sep 2024 - Present

Berkeley, CA

- Working with PhD student on dataset for fine-tuning/RL post-training LLMs to generate better Verilog
- Author on paper accepted for oral presentation at MLArchSys

University Tapeout Project

Jan 2025 - May 2025

- Designed interrupt-based, multi-channel DMA engine for DSP chip on 16nm process
- Wrote 1000+ lines of RTL and constrained random integration tests to verify functionality

Undergraduate Researcher | Berkeley Wireless Research Center

Jan 2024 - Jun 2024

Berkeley, CA

- Designed sensor data streaming hardware on Opal Kelly FPGA board to transfer data in real-time
- Using Verilog and Vivado to convert multi-channel serial input data into custom format based on USB 2.0
- Created real-time receiver with C++ to filter and store incoming 48MHz data

LEADERSHIP & EXTRACURRICULARS

EECS 151 Teaching Assistant | UC Berkeley EECS

Aug 2025 - Present

Berkeley, CA

- Running and developing labs for Berkeley's digital circuit design class
- Led transition to new FPGA boards, setting up new build flows and modules to 2x performance
- Added support for Linux RTOS with UART & shell as extension for RISC-V FPGA softcore final project

President | Berkeleytime

Sep 2023 - Present

Berkeley, CA

- Leading 30+ students to maintain Berkeley's top open source course planning platform, ~30k users
- Lead 5+ subteams per semester to rewrite existing website for better reliability, performance, and features
- Improved member retention rate by 4x, number of applicants by 2x, increased funding by 30x

Administrative Co-ordinator | Residential Life Safety Program

Aug 2023 - Present

Berkeley, CA

- Working with team of 7 to manage 150+ student employees with nightly shifts
- Created spreadsheet/script automations to transform admin processes, significantly simplifying jobs